

Project Title

Urban Heat Island Analysis Using Land Surface Temperature and
Built-up Density

**Project Report Submitted in fulfillment of the Requirments for
the Award of the Internship of Summer Training Program Space
Science Technology**

Subject Name

Remote Sensing & GIS Using Python

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1. TITLE

Urban Heat Island Analysis Using Land Surface Temperature and Built-up Density

2. OBJECTIVE

The main goal of this project is to study the Urban Heat Island (UHI) effect in Pune District, Maharashtra, using satellite based remote sensing data.

Cities tend to be hotter than the rural area around them, mainly because of the concrete buildings, roads, and less vegetation. This difference in temperature can be observed in many cities and is known as the Urban Heat Island (UHI) effect. To study this, Landsat 8 satellite imagery was used to calculate three things:

- **Land Surface Temperature (LST):** How hot is the ground surface is in different parts of Pune.
- **Normalized Difference Vegetation Index (NDVI):** How much green vegetation is present.
- **Normalized Difference Built-up Index (NDBI):** How much of an area is covered by buildings and roads.

By comparing these three outputs, the project tries to understand which zones of Pune are the hottest, whether these zones are also the most built-up, and how vegetation helps in keeping temperature lower.

The tools used are:

- **Google Earth Engine:** Used for satellite data processing and analysis.
- **Python:** Used to generate graphs and statistical analysis.
- **QGIS:** Used to prepare final thematic map layout.

3. STUDY AREA

3.1 Location

The study area is Pune District, located in the state of Maharashtra, India. Pune is one of the fastest growing cities in India. In the last two decades, the district has experienced rapid urban expansion due to increase in population, industrial development, and infrastructure growth. Such rapid urban growth makes Pune District an appropriate study area for analysing the Urban Heat Island (UHI) effect.

Pune District has densely built up urban regions, agricultural areas, water bodies, and hilly areas. This diversity in land helps in understanding spatial variations in surface temperature across the district.

Parameter	Details
Country	India
State	Maharashtra
District	Pune
Area	Approx 15,643 km ²

3.2 Area of Interest (AOI) Map

The Pune District administrative boundary was obtained from the GADM database and imported into Google Earth Engine as an asset. This boundary was then used to define the Area of Interest (AOI) and clip all satellite derived outputs to the study area.



Figure 1. Area of Interest (AOI) of Pune District used for analysis in GEE.

4. DATA USED

For this project, one main satellite dataset was used, which provided all the information needed for calculating LST, NDVI, and NDBI. Using a single satellite sensor simplified the workflow and ensured that all the required bands had the same spatial resolution, avoiding need for additional resampling.

4.1 Primary Dataset

Parameter	Details
Satellite	Landsat 8 (OLI/TIRS)
Collection	Collection 2, Tier 1, Level 2 Surface Reflectance
GEE Dataset ID	LANDSAT/LC08/C02/T1_L2
Spatial Resolution	30 meters
Temporal Resolution	16 days
Analysis Period	1 March 2024 to 31 May 2024
Cloud Cover Filter	Less than 10%
Source	Google Earth Engine Data Catalog

4.2 Bands Used

Band	Wavelength	Purpose
Band 4 (Red)	0.64 – 0.67 μm	NDVI calculation
Band 5 (NIR)	0.85 – 0.88 μm	NDVI & NDBI calculation
Band 6 (SWIR1)	1.57 – 1.65 μm	NDBI calculation
Band 10 (Thermal)	10.6 – 11.19 μm	Land Surface Temperature

4.3 Why Summer 2024?

March, April, and May are the hottest months in Pune. The Urban Heat Island effect is strongest during these months. If we use data from entire year, then monsoon and winter would bring the temperatures down and make the UHI signal weaker than it actually is. Therefore, using a summer composite provides a clearer representation of the Urban Heat Island effect.

4.4 AOI Boundary

Parameter	Details
Boundary Level	District Level (Administrative Level 2)
Source	Global Administrative Areas(GADM)
URL	https://gadm.org
Format	Shapefile imported into GEE as Asset

5. METHODOLOGY

The project was carried out in three stages:

- Google Earth Engine for satellite processing.
- Python for statistical analysis
- QGIS for final cartographic maps.

All the tools are part of the ISA internship curriculum.

5.1 GEE Workflow

Step 1: Setting up GEE

Google Earth Engine was used as the main platform for satellite data processing. It allowed the required Landsat datasets to be accessed and processed directly through a web based interface. I opened GEE code editor at <https://code.earthengine.google.com>.

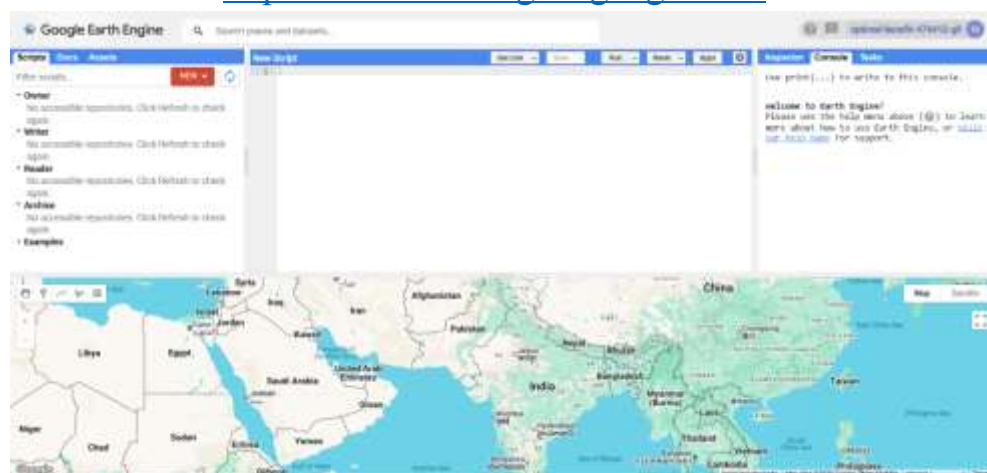


Figure 2. Google Earth Engine Code Editor.

Step 2: Loading Landsat 8 and Filtering

The Landsat 8 Collection 2 level 2 dataset was selected in GEE for analysis. Images corresponding to the Pune District study area were obtained for the summer period from 1 March 2024 to 31 May 2024. To improve the quality of the analysis, only images with less than 10% cloud cover were considered. The selected images were then combined to create a summer dataset for Pune District, which was

further used for calculating NDVI, NDBI, and Land Surface Temperature.

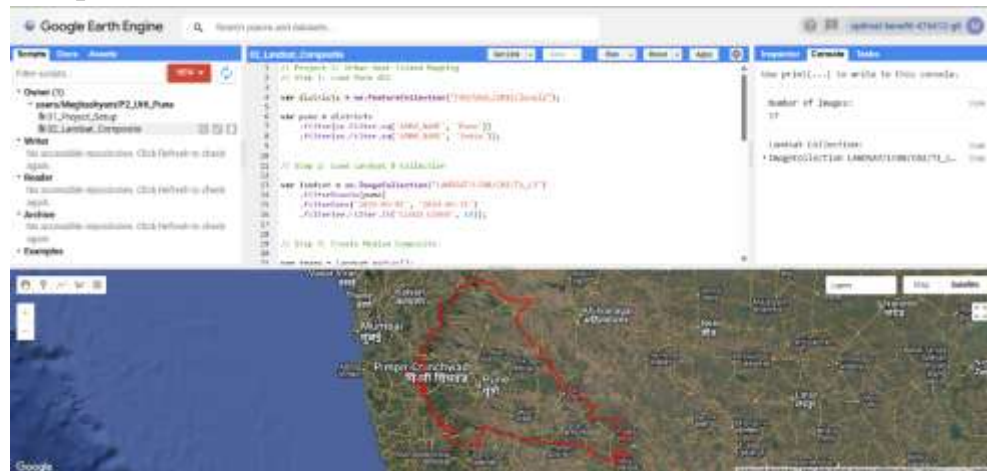


Figure 3. Landsat 8 images selected for Pune District after applying date range & cloud cover filters in GEE.

Step 3: True Colour Composite

A true colour composite was added to the map to visually check that the correct data was loaded and the AOI was covered properly.

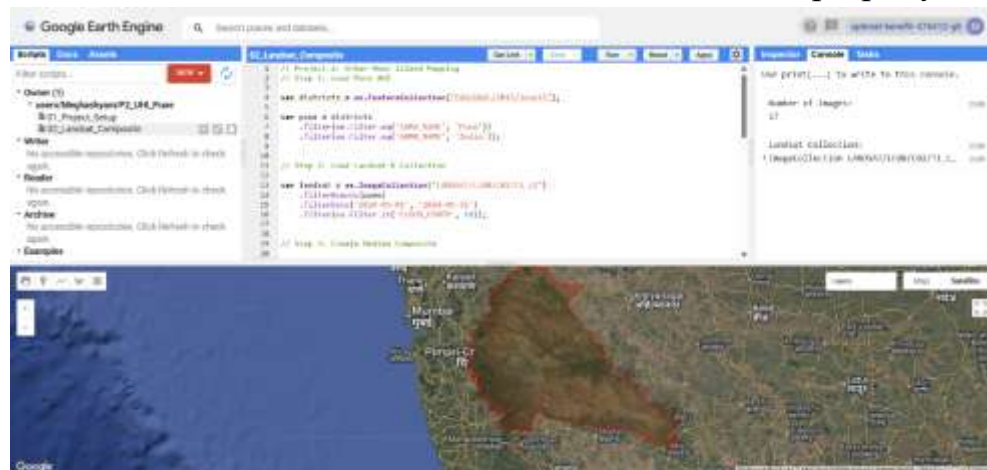


Figure 4. True Colour Composite of Pune District, Summer 2024.

Step 4: Calculating NDVI

NDVI was calculated using the normalized difference between the Near Infrared (Band 5) and Red (Band 4) bands. Higher NDVI values indicate denser vegetation, while lower values represent built up surfaces.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

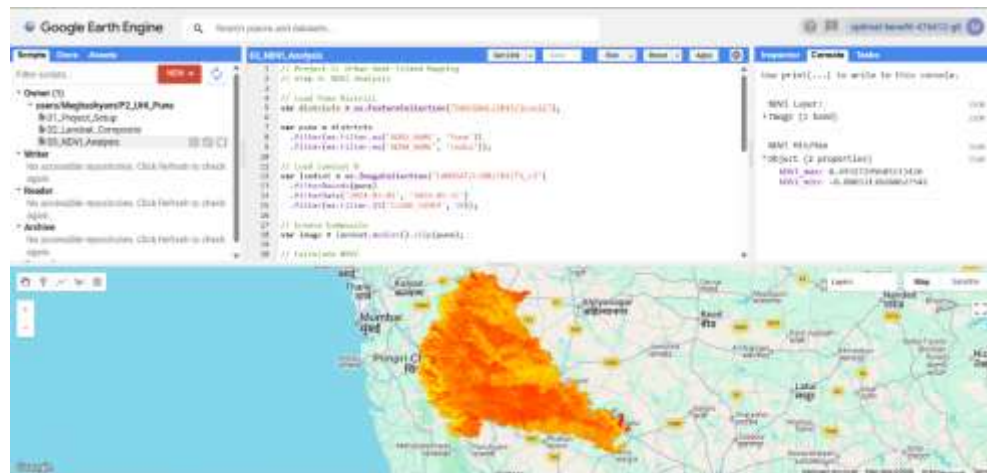


Figure 5. NDVI Layer of Pune District in GEE.

Step 5: Calculating NDBI

NDBI was derived using the SWIR1 and NIR bands to identify built up areas. Higher NDBI values represent more built up surfaces like buildings and roads.

$$\text{NDBI} = (\text{SWIR1} - \text{NIR}) / (\text{SWIR1} + \text{NIR})$$

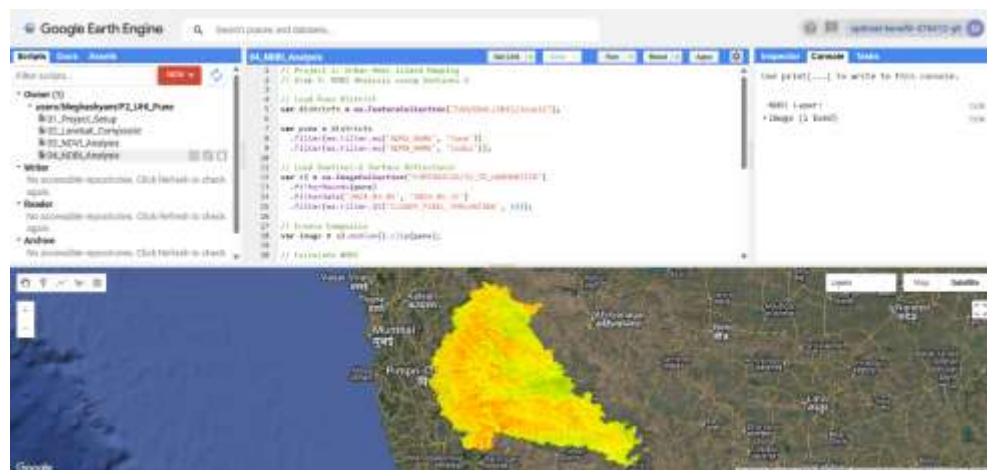


Figure 6. NDBI Layer of Pune District in GEE.

Step 6: Calculating LST

LST was derived from Landsat 8 Band 10 (Thermal Infrared). The Landsat Collection 2 level 2 product provides thermal values as scaled digital numbers. To convert these to land surface temperature

in Celsius, the following USGS provided conversion formula was applied:

$$\text{LST}(\text{°C}) = (\text{Band } 10 \times 0.00341802) + 149 - 273.15$$

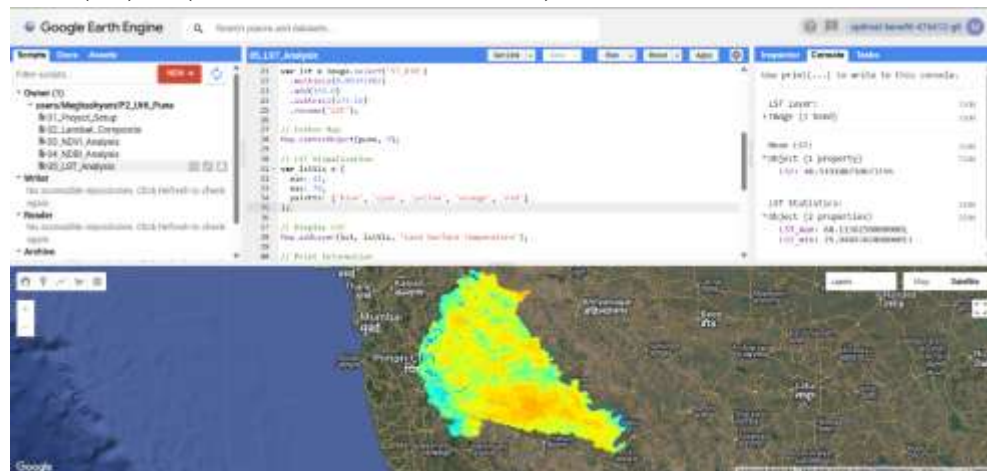


Figure 7. LST Layer of Pune District in GEE.

Step 7: Creating UHI Zones

Based on the LST values, Pune District was classified into five Urban Heat Island categories ranging from Very Low to Very High intensity.

UHI Class	Temperature Range
Very Low	Less than 28°C
Low	28°C to 32°C
Moderate	32°C to 36°C
High	36°C to 40°C
Very High	Above 40°C

The temperature thresholds were selected based on the observed summer LST distribution over Pune District to distinguish relatively cooler and hotter zones for comparative UHI assessment.

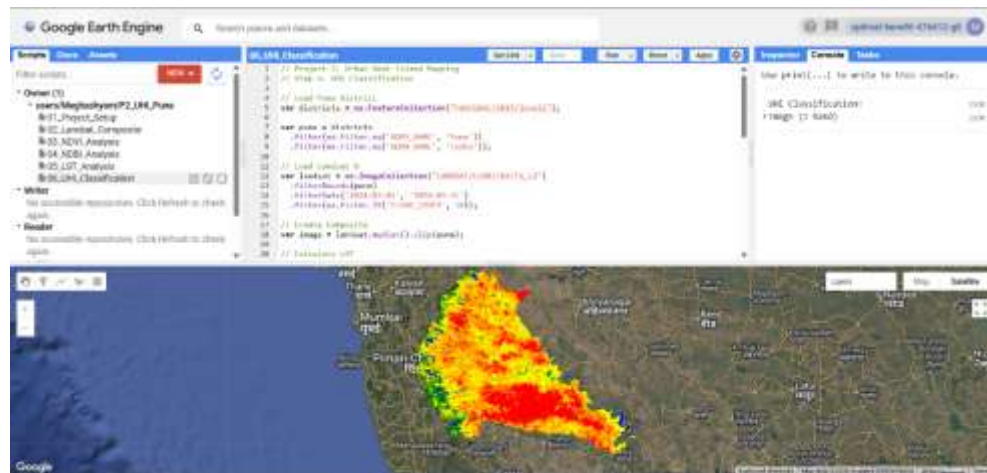


Figure 8. UHI Zone Classification of Pune District in GEE.

Step 8: Exporting to Google Drive

All four rasters were exported to Google Drive using the GEE tasks panel for further processing in QGIS and Python.

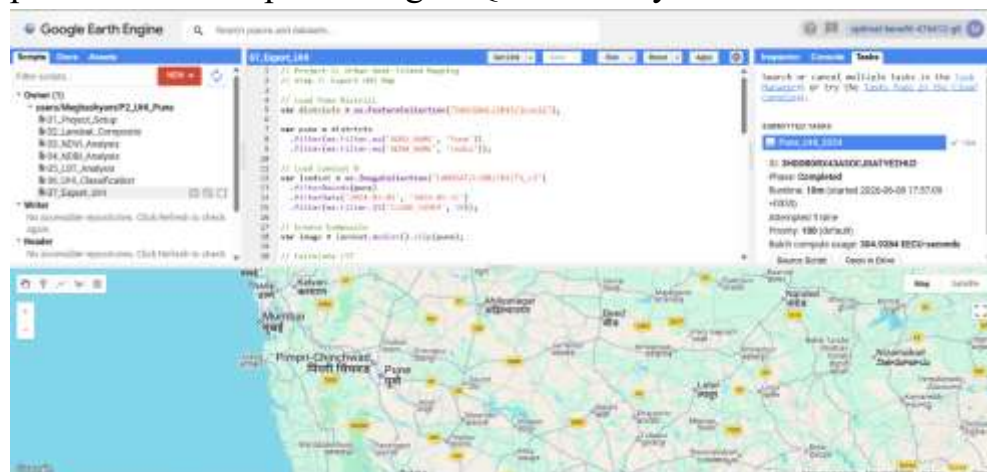


Figure 9. GEE tasks panel showing export jobs running.

5.2 Python Analysis Workflow

After the raster exports were downloaded from Google Drive, all statistical analysis and graph generation was done locally in VS Code using Python. The libraries used were Rasterio, NumPy, Pandas, Matplotlib, and SciPy.

The exported TIFF files (LST, NDVI, NDBI) were read directly using Rasterio and converted into NumPy arrays. Matplotlib was

used to create all graphs, and SciPy was used for correlation calculation.

Step 9: Reading Raster Files

The exported raster datasets were processed in Python using Rasterio and NumPy. Pixel values were extracted, and invalid values were removed before performing statistical analysis.

Step 10: LST Histogram

A histogram was generated to show the overall distribution of land surface temperature across Pune District.

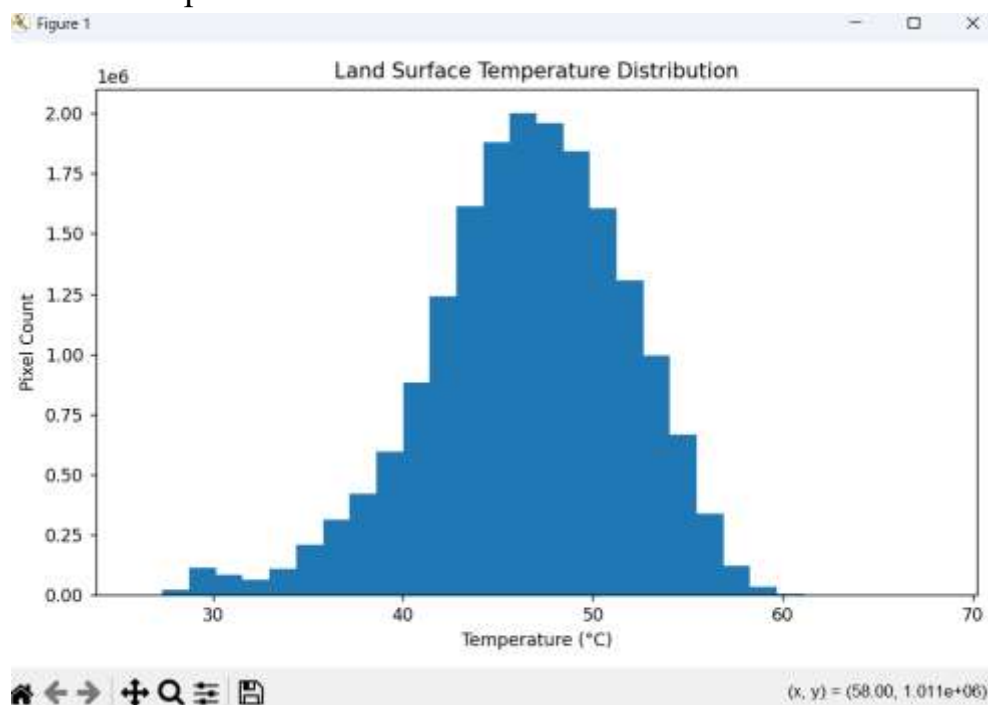


Figure 10. Histogram showing distribution of LST values across Pune District.

Step 11: UHI Class Distribution Chart

A bar chart was generated to show what percentage of the district falls under each UHI temperature class.

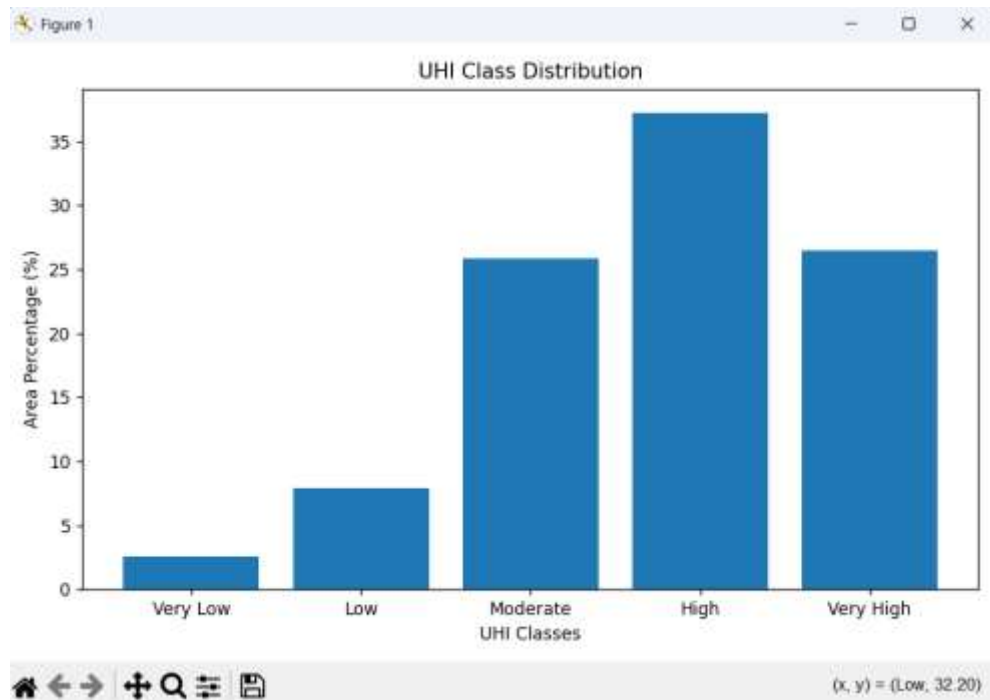


Figure 11. Bar chart showing area percentage of each UHI class.

Step 12: Scatter Plot LST vs. NDBI

A scatter plot with a regression trend line was created to study the relationship between built up density and surface temperature.

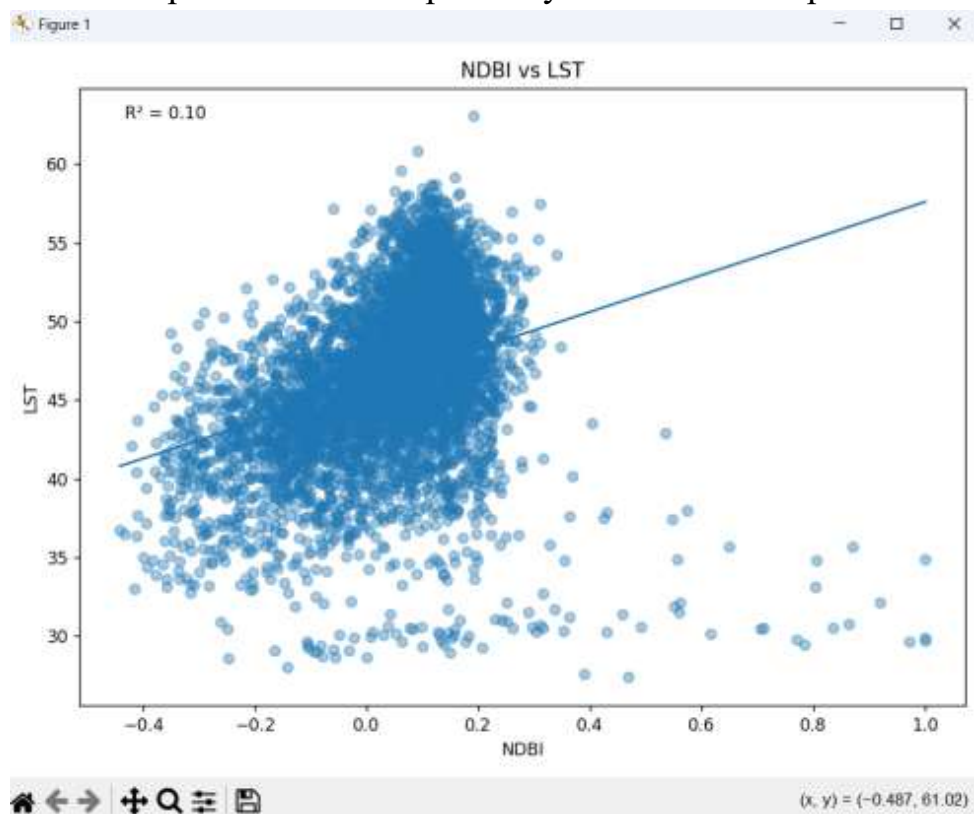


Figure 12. Scatter Plot: NDBI vs. LST

Step 13: Scatter Plot LST vs. NDVI and Correlation

A scatter plot and Pearson correlation coefficient were used to investigate the relationship between NDVI and LST.

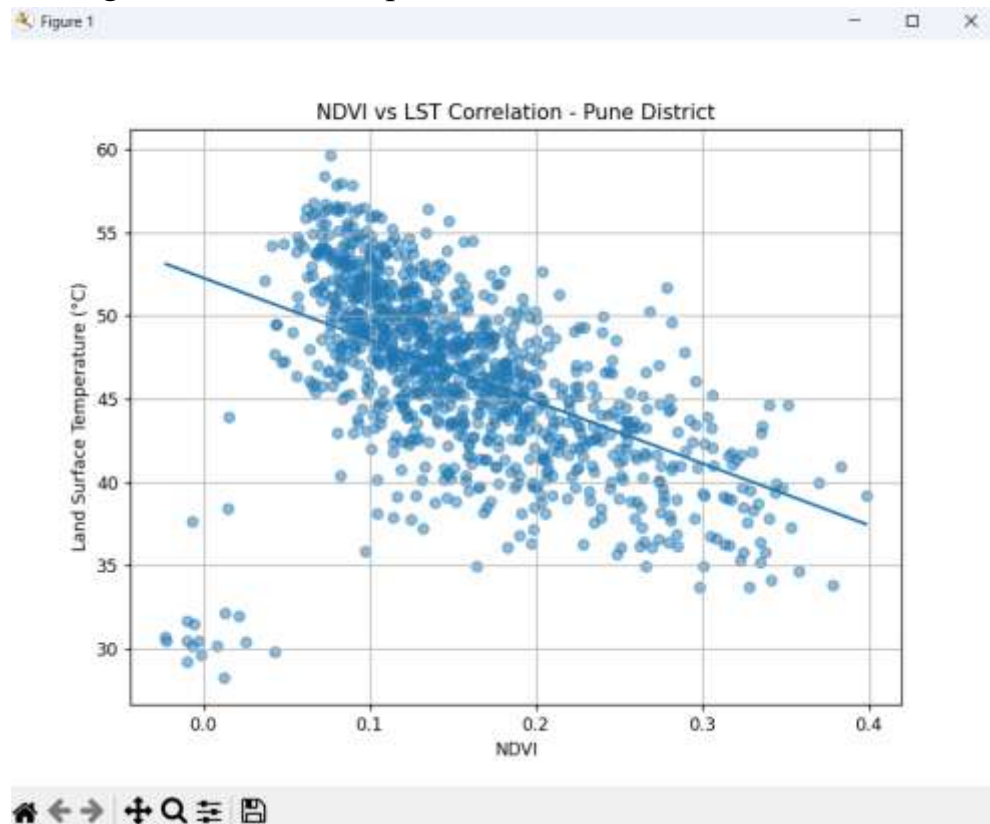


Figure 13. Scatter Plot: NDVI vs. LST

5.3 QGIS Workflow

QGIS was used only for map preparation and visualization. No analytical calculation like NDVI, NDBI, or LST were done in QGIS. It was used to load the final raster outputs, apply symbology, and prepare the print layout.

Step 14: Loading Rasters into QGIS

The exported raster datasets were loaded into QGIS for visualization and layout preparation.

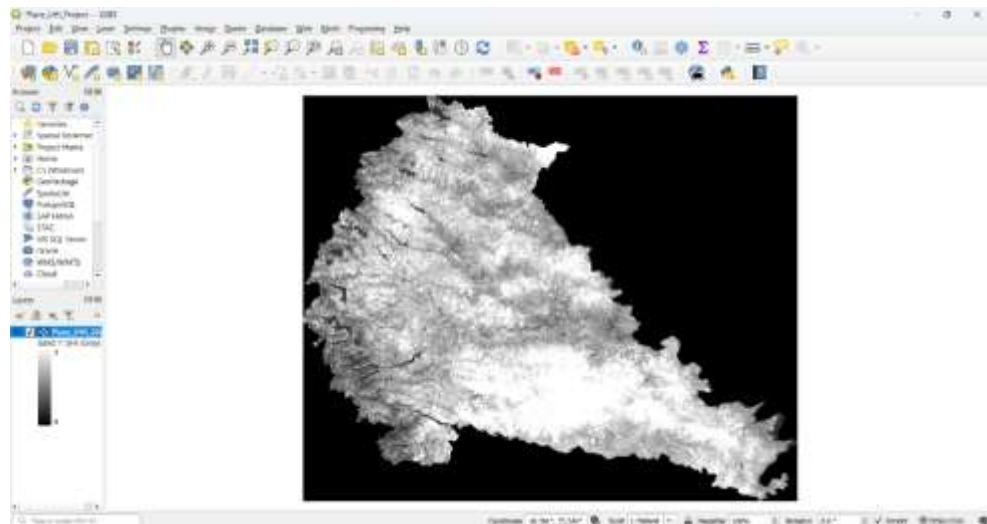


Figure 14. All raster layers loaded in QGIS.

Step 15: Applying Symbology

The UHI Zones raster was styled using 5 class colour scheme consisting of Blue, Green, Yellow, Orange, and Red to represent Very Low, Low, Moderate, High, and Very High UHI intensity zones respectively. The symbology was applied through the Layer Properties panel in QGIS. White areas in the legend represent No Data.

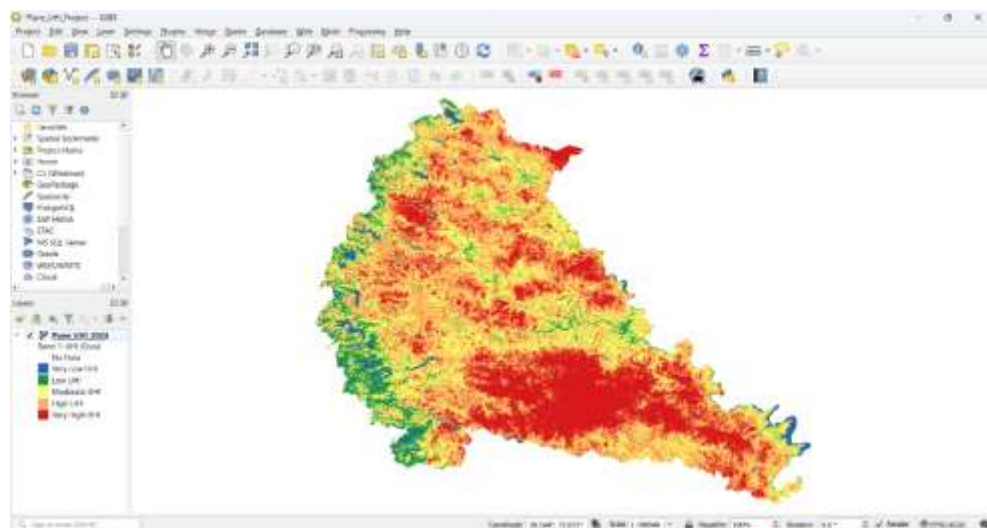


Figure 15. UHI Zones layer with colour styling applied in QGIS.

Step 16: Designing the Print Layout

The QGIS print layout was used to arrange the map and add a title, legend, scale bar, north arrow, and insert map showing Pune's location within Maharashtra.

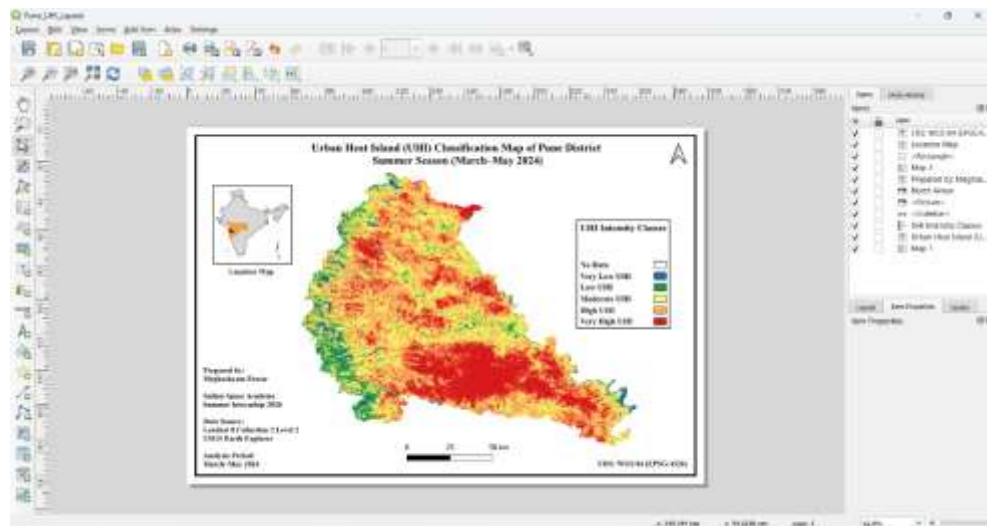


Figure 16. Print Layout design with map elements in QGIS.

Step 17: Exporting the Final Map

The layout was exported as a 300 DPI PNG image for inclusion in the report.

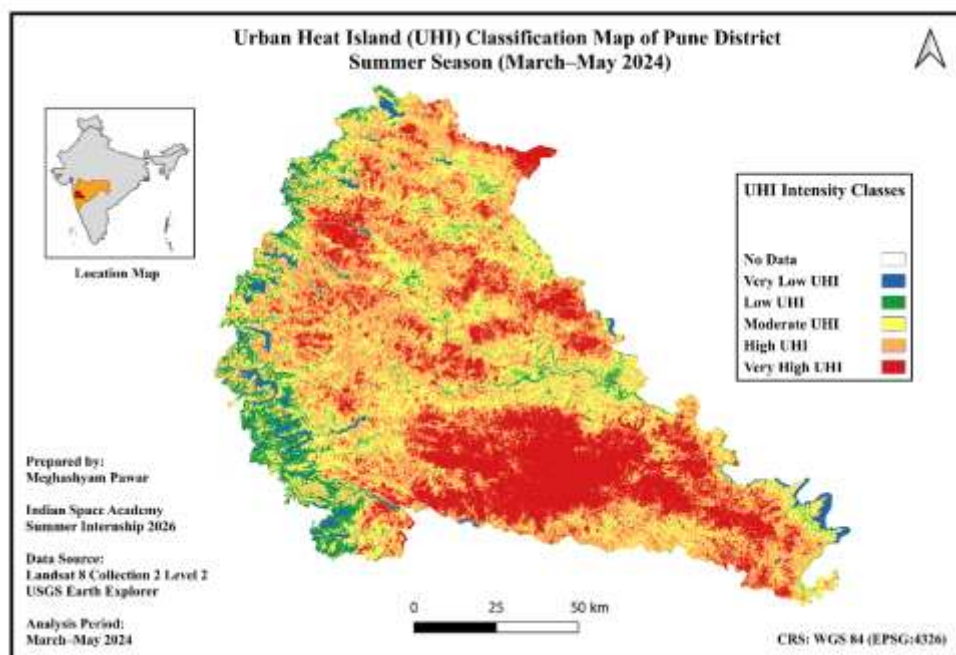


Figure 17. Final Urban Heat Island Map, Pune District, Summer 2024.

6. RESULTS

Summary statistics of LST, NDVI, NDBI values for Pune District:

Parameter	Minimum	Maximum	Mean
LST (°C)	29.05	57.87	44.78
NDVI	-0.045	0.492	0.171
NDBI	-0.578	1.000	0.010

6.1 Land Surface Temperature

The LST map indicates a clear temperature difference across Pune District. Areas with dense urban development showed noticeable higher surface temperature. In contrast, forested hills, agricultural regions, and water bodies generally recorded lower temperatures.

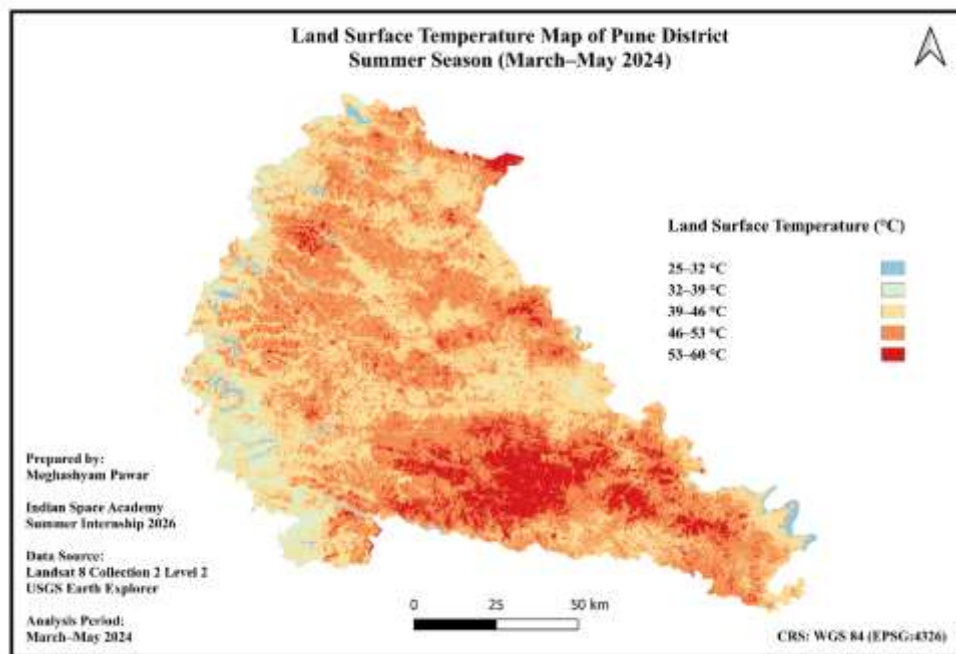


Figure 18. LST Map, Pune District (March – May 2024).

6.2 Vegetation Cover (NDVI)

The NDVI map indicates that vegetation cover is mostly concentrated in the hilly regions, agricultural areas, and other parts of Pune District. Highly urbanised areas showed lower NDVI values, indicating limited vegetation cover.

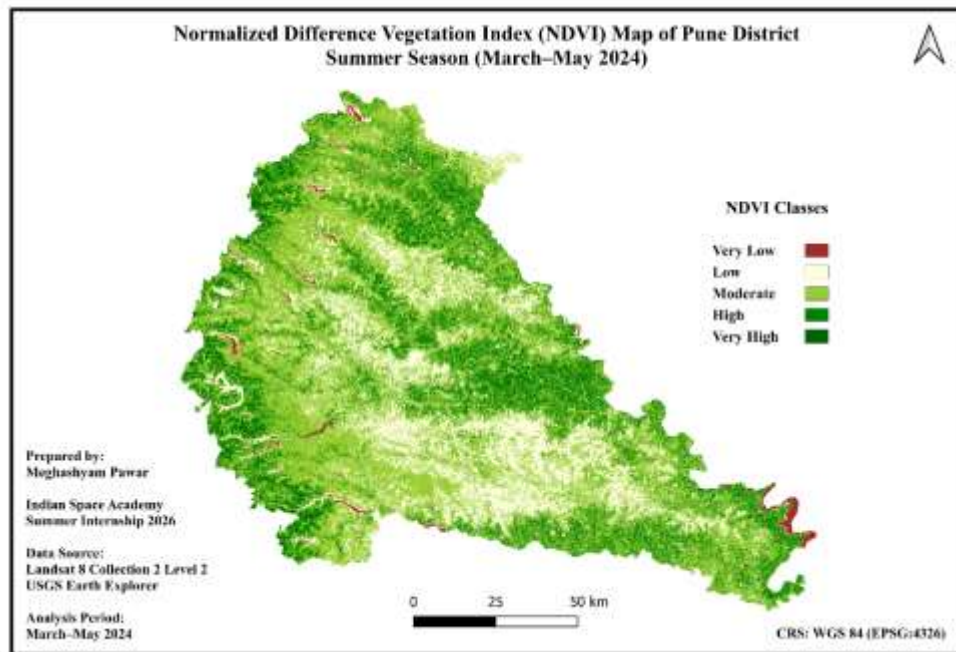


Figure 19. NDVI Map, Pune District (March – May 2024).

6.3 Built-up Density (NDBI)

The NDBI map highlights regions with a greater concentration of built-up surfaces. These areas showed a noticeable overlap with regions having higher LST values, suggesting a relationship between urban development and increased surface temperature.

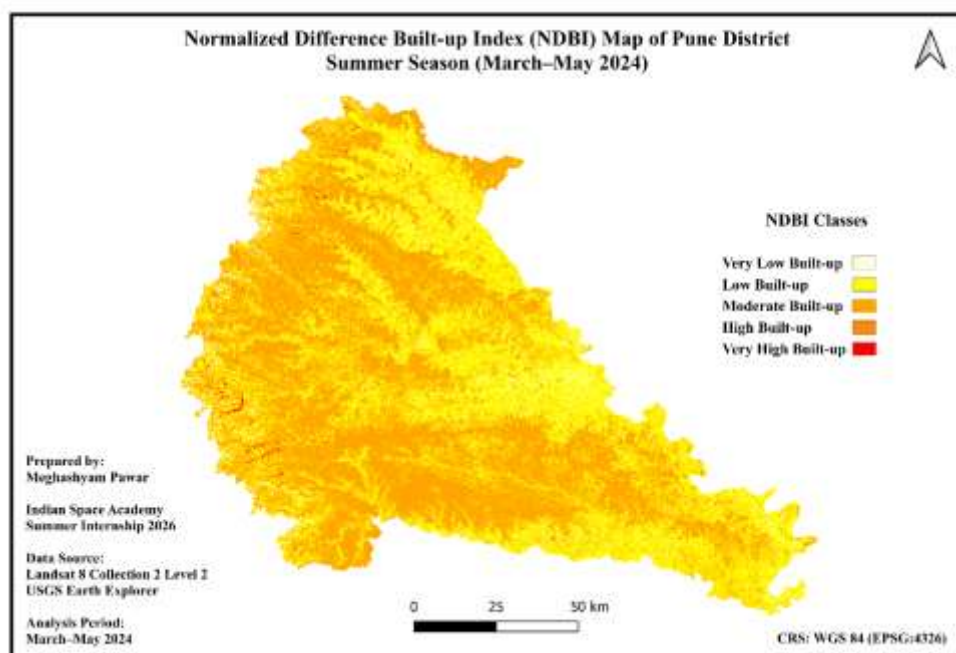


Figure 20. NDBI Map, Pune District (March – May 2024).

6.4 UHI Zone Classification

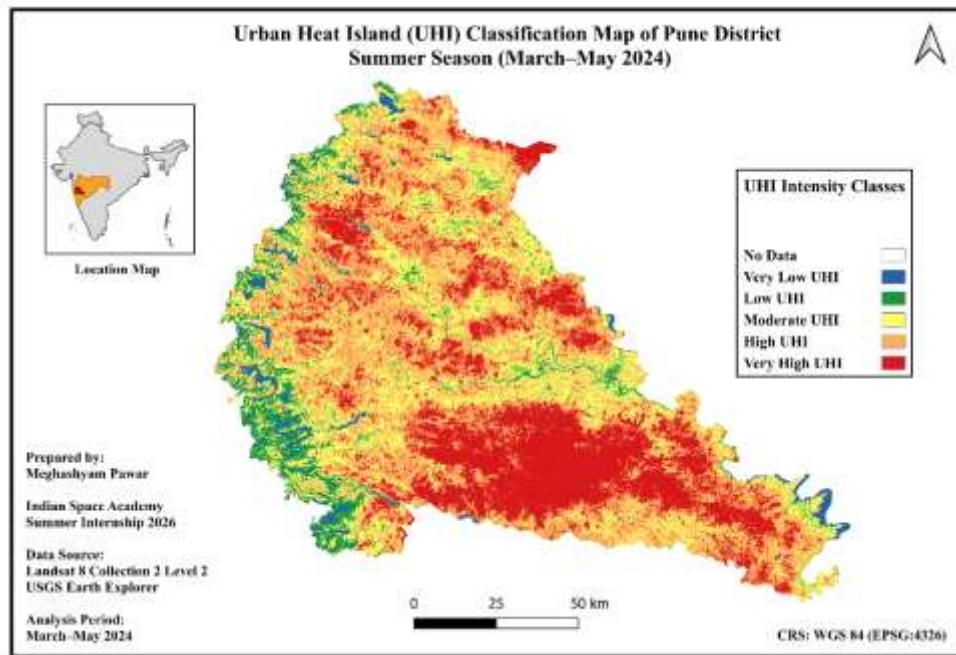


Figure 21. Final UHI Zone Classification Map, Pune District, Summer 2024.

The results of the UHI zone classification are shown in the table below:

UHI Class	Temperature Range	Area (%)
Very Low	Less than 28°C	2.55%
Low	28°C to 32°C	7.9%
Moderate	32°C to 36°C	25.83%
High	36°C to 40°C	37.25%
Very High	Above 40°C	26.47%

The results indicate that more than 63% of Pune District falls under the “High” or “Very High” UHI category during summer. Only about 10% of the area is in the cooler categories, which are mostly the forested hills and water bodies at the edge of the district. These observations suggest that heat stress is widespread across the district during the peak summer season.

6.5 Correlation Analysis

The scatter plot for NDBI vs. LST shows a positive relationship, areas with higher built up index values tend to also show higher surface temperatures. The analysis suggests that highly urbanised zones in Pune experience greater heat compared to less developed areas.

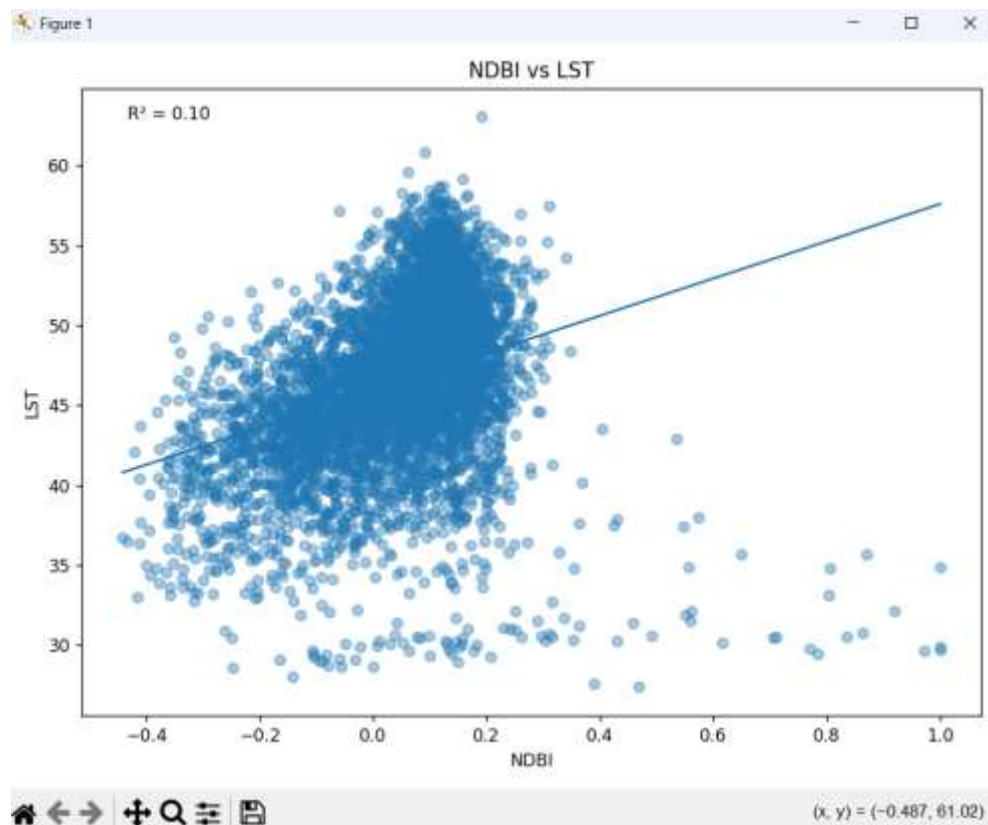


Figure 22. Scatter Plot: NDBI vs. LST

The scatter plot for NDVI vs. LST shows a negative relationship. The Pearson correlation coefficient calculated using SciPy was: $r = -0.4997$, indicating a moderate negative relationship between vegetation cover and surface temperature. The results suggest that areas with higher vegetation cover generally showed lower land surface temperatures. This highlights the role of green cover in reducing surface heat in urban areas.

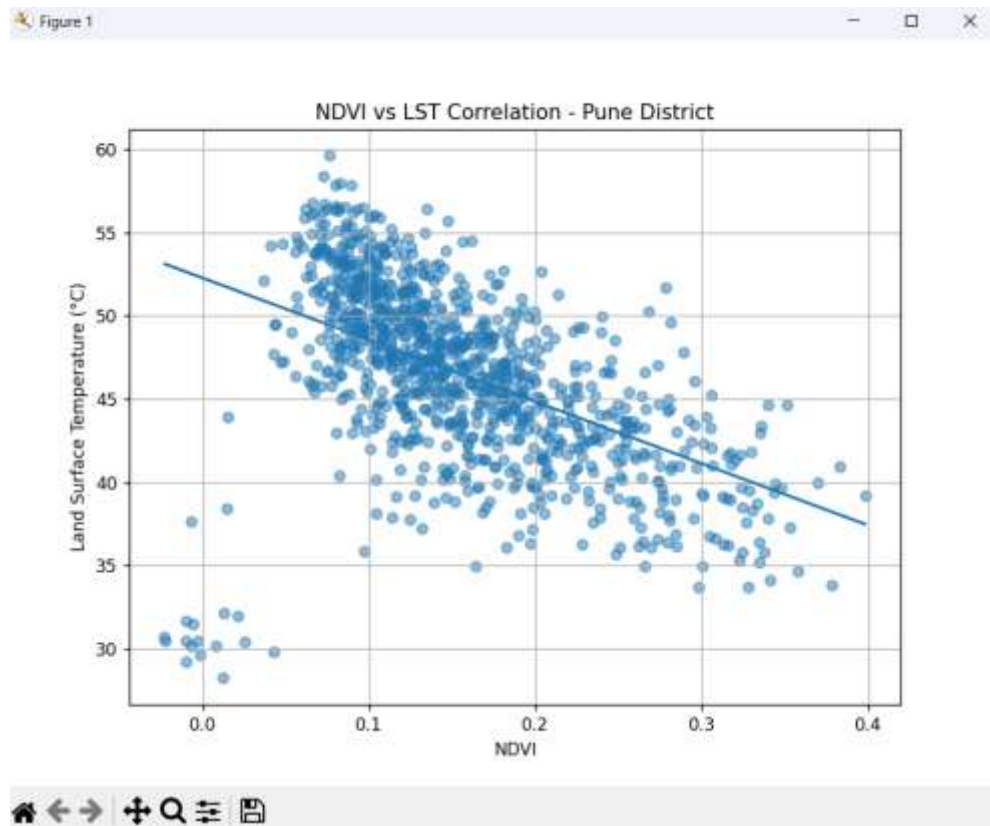


Figure 23. Scatter Plot: NDVI vs. LST

7. CONCLUSION

This project successfully mapped the Urban Heat Island (UHI) effect in Pune District using Landsat 8 satellite data. The satellite data were processed in Google Earth Engine, analysed in Python using Rasterio, NumPy, and SciPy, and the final thematic maps were prepared in QGIS.

The findings indicate that the dense urban areas in central and western parts of Pune experience much higher surface temperature compared to the vegetated outskirts. The analysis showed that more than 63% of the district falls under the “High” or “Very High” UHI categories during summer. The NDVI – LST correlation analysis produced a Pearson correlation coefficient of $r = -0.4997$, indicating that areas with higher vegetation cover generally experienced lower surface temperatures. In contrast, the positive relationship observed between NDBI and LST suggests that built-up areas contribute to increased surface heating.

The result shows that Pune’s rapid urbanization had a measurable effect on its local surface temperature patterns. The findings also indicate that increasing green cover in the high UHI zones could help in reducing surface heat in those areas.

Limitations:

- Landsat 8’s 30-meter spatial resolution is not enough to separate small features like narrow streets or small parks.
- This study was limited to the summer season of 2024. A multi-year analysis would provide a clearer understanding of how UHI patterns have changed over time with urban expansion.
- Ground-level temperature data from meteorological stations was not available to verify the satellite-derived LST values in this study.
- Satellite derived Land Surface Temperature represents surface conditions rather than actual air temperature experienced by people. Therefore, the results should be interpreted as surface heat patterns and not direct measurements of ambient air temperature.

Future Improvements:

- Comparing 2014 vs. 2024 LST data to study growth of UHI alongside urban expansion in Pune.
- Overlaying population density data to identify which communities are most affected by high surface temperatures.

8. REFERENCES

- Google Earth Engine: <https://earthengine.google.com/>
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- GADM. Global Administrative Areas Database: <https://gadm.org/>
- Rasterio. Access to geospatial raster data in python: <https://rasterio.readthedocs.io/en/stable/>
- NumPy. Numerical computing with Python: <https://numpy.org/>
- SciPy. Scientific computing tools for Python: <https://scipy.org/>
- Matplotlib. Visualization with Python: <https://matplotlib.org/>
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